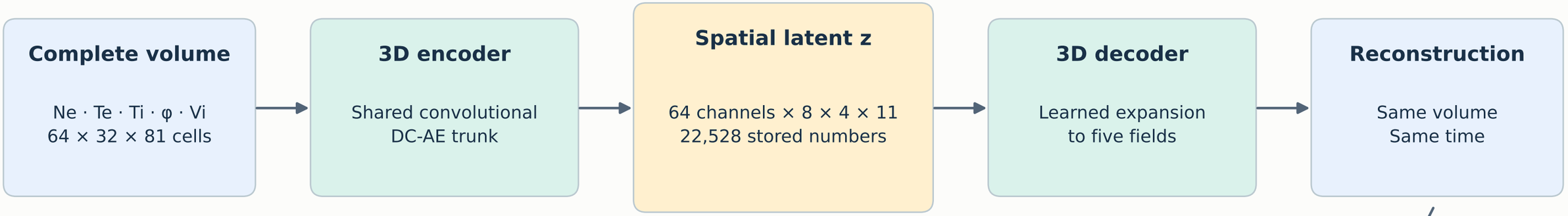
01 / Full-volume reconstruction

Question: what survives compression when the model sees the complete state?



Before encoding: native toroidal grid 81 → 88; log-density transform and fitting-only normalization.

After decoding: inverse transforms and toroidal grid 88 → 81; score in physical fields on native cells.

Training: encoder and decoder learn together

½ normalized mean error + ½ normalized fluctuation error
Five fields, native volume weights, boundary supervision

Mean = instantaneous toroidal n=0. Fluctuation = full field minus that mean.

The latent is a grid of learned features, not 64 physical variables. No input bypasses the bottleneck.

Reference: balanced boundary-supervised shared codec. Flux, spectra and cross-phase are evaluated separately.